INCH-POUND

MS21254R
10 January 2001
SUPERSEDING
MS21254P
30 October 1999

MILITARY SPECIFICATION SHEET

EYE END, TURNBUCKLE, CLIP LOCKING

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-DTL-8878, QPL-8878.

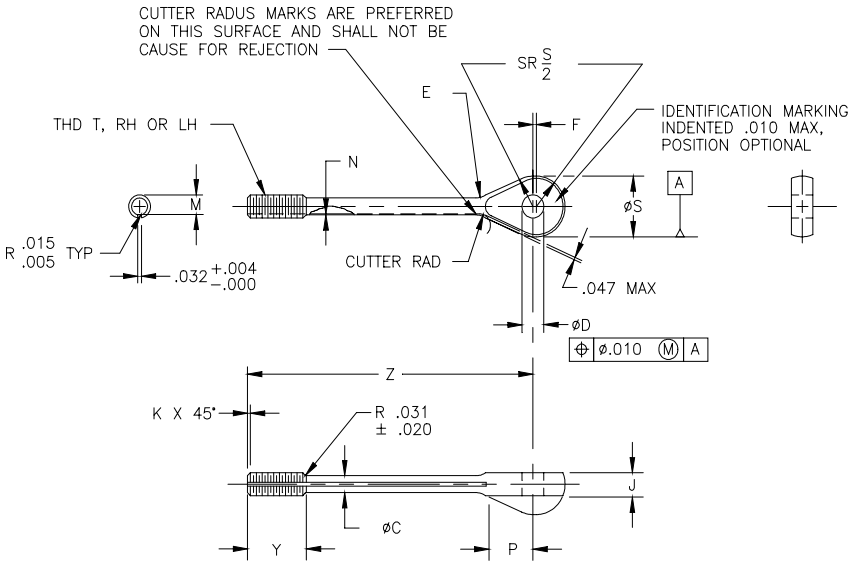

FIGURE 1. Eye end, turnbuckle, clip locking.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:29Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions.

Dash number		Wire rope diameter		Minimum breaking strength lb	Thread T UNF-3A	∅C +.006 -.000	∅D +.002 -.000	E ±.031 radius	F +.010 -.000	J ±.005
Direction of thread										
RH	LH	Nominal reference	Minimum							
-2RS	-2LS	1/16	.062	800	.1380 (#6)-40	.094	.190	.125	.031	.125
-2RL	-2LL									
-3RS	-3LS	3/32	.093	1,600	.190 (#10)-32	.133		.156		.188
-3RL	-3LL									
-5RS	-5LS	5/32 1/	.156	3,200	.250 (1/4)-28	.189	.250	.219	.047	.219
-5RL	-5LL									
-6RS	-6LS	3/16	.187	4,600	.3125 (5/16)-24	.243	.313	.250		.281
-6RL	-6LL									
-8RL	-8LL	1/4	.250	8,000	.3750 (3/8)-24	.306	.375	.281	.063	.328
-9RL	-9LL	9/32	.281	12,500	.4375 (7/16)-20	.362	.438	.359	.078	.375
-10RL	-10LL	5/16	.312	17,500	.5000 (1/2)-20	.425	.500	.406		.469

1/ For .125 (1/8) inch diameter wire rope use -5 parts, minimum breaking strength 2,200 lb (ref).

TABLE I. Dash numbers and dimensions - Continued.

Dash number		Wire rope diameter		K +.000 -.015	M		N	P ± .031	∅S +.025 -.010	Y ± .047	Z +.031 -.015
Direction of thread											
RH	LH	Nominal reference	Minimum		Maximum	Minimum	Maximum				
-2RS	-2LS	1/16	.062	.031	.1139	.1094	.0096	.325	.375	.375	1.500
-2RL	-2LL										2.375
-3RS	-3LS	3/32	.093		.1638	.1568	.0077	.416	.500	.500	1.625
-3RL	-3LL										2.500
-5RS	-5LS	5/32	.156		.2224	.2152	.0103	.512	.625	.625	1.750
-5RL	-5LL										2.625
-6RS	-6LS	3/16	.187	.047	.2830	.2754	.0064	.527	.688	.750	1.875
-6RL	-6LL										2.750
-8RL	-8LL	1/4	.250		.3454	.3378	.0057	.678	.875	.875	2.875
-9RL	-9LL	9/32	.281		.4052	.3972	.0056	.831	1.063	1.000	3.375
-10RL	-10LL	5/16	.312	.063	.4678	.4597	.0058	.900	1.188		3.625

REQUIREMENTS:

1. Material: 17-4PH stainless steel shall be in accordance with SAE AMS 5643 or ASTM A 564. Steel, carbon or alloy shall be in accordance with MIL-DTL-8878.

2. Protective treatment: Protective treatment shall be in accordance with MIL-DTL-8878.

3. Heat treatment: Steel, carbon or alloy shall be heat treated to 125 to 145 KSI in accordance with SAE AMS-H-6875. 17-4PH shall be in accordance with MIL-DTL-8878.

4. Finish: Finish shall be in accordance with MIL-DTL-8878.

5. Threads: Threads shall be in accordance with FED-STD-H28/20.

6. Tolerances: Unless otherwise specified, tolerances: decimals ± .010, angles ± .5°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:29Z
Check the source to verify that this is the current version before use.

7. Outside diameter (O.D.): O.D. of the "F" dimension may be a flat area.

8. Part or identifying number (PIN): The PIN shall consist of the basic MS number, followed by a dash number; C in lieu of dash indicates 17-4PH (dash indicates carbon or alloy steel); first letter following dash number or letter C indicates direction of thread (left or right hand), second letter following dash number indicates length (short or long).

Example of PIN:

MS21254C3LS - Eye, turnbuckle, 17-4PH, .1900 (#10)-32 left hand thread, short.
MS21254-3RL - Eye, turnbuckle, steel, .1900 (#10)-32 right hand thread, long.

NOTES:

1. Remove all burrs and sharp edges. (See procurement specification for definition.)

2. Dimensions are in inches.

3. Interpret drawing in accordance with ASME Y14.5M.

4. MS21254 items can replace AN165 items of like thread size and material.

5. For .125 (1/8) inch diameter wire rope use -5 parts, minimum breaking strength 2,200 pounds (ref).

6. Steel, carbon and alloy eye end turnbuckles are inactive for new design. 17-4PH stainless steel eye end turnbuckles shall be used for new design and can be used to replace comparable steel parts.

7. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

8. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.

9. For clip locking of turnbuckles, see MS33736.

10. The parts covered by dash numbers shown on AN165 are canceled after 25 November 1963. Use the dash numbers given in table I. The canceled parts cannot replace the new parts and should be used until existing stock is depleted. Use only the MS21254 parts for new design and substitution. Substitution shall be in accordance with table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:29Z
Check the source to verify that this is the current version before use.

TABLE II. Substitution table.

PART NUMBER	
Canceled part AN165	New part MS21254
Dash number	Dash number
8	2
16	3
22	5
32	5
46	6
61	8
80	8
125	9
175	10

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Reviewers:
Army - CR4
Navy - MC
Air Force - 82

Preparing activity:
DLA - GS5

(Project 1560-0282)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:29Z
Check the source to verify that this is the current version before use.